

Umang Garg

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📍 U.C. Santa Barbara, CA

RESEARCH PROFILE

AI Research Intern Candidate — Physics-Informed Generative Modeling, Computer Vision, Diffusion Models, Scientific ML

5th-year Ph.D. researcher at UCSB Vision Lab developing physics-informed models for scientific imaging, including symmetry-group-aware EBSD super-resolution and conditional latent diffusion for 3D/4D microstructures; broader work spans medical image segmentation, trajectory foundation models, and adversarial robustness and training.

TECHNICAL SKILLS

ML/AI: PyTorch, latent diffusion, Transformers, Segment Anything, CNNs, GNNs, self-supervised learning, multimodal foundation models, adversarial training

Scientific/Imaging: super-resolution, crystal orientation maps, 3D/4D microstructure generation, medical image segmentation, trajectory modeling

Programming/Tools: Python, C/C++, PyTorch DDP, MATLAB, TensorFlow, OpenCV, Git, Linux, Docker/containers, Dream3D

EDUCATION

University of California, Santa Barbara (UCSB)

5th year Ph.D. in Computer Engineering, Advisor: Prof. Manjunath (Vision lab)

Sept 2021 – Present

Santa Barbara, CA

Birla Institute of Technology and Science, Pilani

B.E. in Electronics and Instrumentation

Aug 2016 – May 2020

Pilani, India

RESEARCH PROJECTS

SG-SRAN: Symmetry-Aware EBSD SR  | [PyTorch](#), [e3nn](#)

Jan '25 - Current

- Developed SG-SRAN, a symmetry-aware 4×4 EBSD super-resolution model for FCC and HCP crystals; first-author manuscript under review at Nature Machine Intelligence.
- Encoded $SO(3)/G$ orientations with Reynolds projection and routed high-resolution subpixels before dictionary decoding; **49K FCC / 27K HCP parameters**.
- Achieved **0.0076 / 0.0079 rad** FCC/HCP round-trip errors and zero-shot evaluation on unseen alloys across five seeds.

Low-Latency Single-Step Diffusion for Point-Cloud Render Cleanup  | [Diffusion](#) **Jun '26 – Current**

- Developing a single-step diffusion model for direct point-cloud render cleanup, targeting low-latency removal of Gaussian-splatting / rendering artifacts.
- Exploring multimodal drone-image fusion extending Diffix+ techniques and multi-view grounding for streaming low-latency frame-consistent novel-view synthesis.

GPS-Trajectory Foundation Model (GPS-MTM)  | [PyTorch DDP](#)

Aug '24 - Sept '25

- **“First-author preprint”** Trained a masked trajectory foundation model with PyTorch DDP on terabyte-scale GPS data for anomaly detection and trajectory reasoning; curated an expanded GeoLife POI dataset.

Enhanced SAM-adapted MRI-CT segmentation model  | [PyTorch](#), [Python](#)

Jun '24 - Sept '24

- **“Neurosurgery submitted”** Adapted Segment-Anything model for 7-region MRI/CT brain segmentation for normal-pressure hydrocephalus analysis detection ([clinically used by neurosurgeons](#))
- **11%** and **25%** dice score improvement in ventricles, subarachnoid regions

Adversarial ML attack and defense design (CS 291A)  | [PyTorch](#), [Python](#)

Jan '23 - Mar '23

- Crafted FGSM, PGD attack on ResNet-18 for image recognition CIFAR-10.
- Robustified accuracy from **0-62%** using adversarial training with TRADES and semi-supervised learning.

PATENTS AND PUBLICATIONS

Manuscripts / Preprints

- [1] **First author.** Umang Garg, Bowen Zhang, Anantajit Subrahmanya, Chandrakanth Gudavalli, and B. S. Manjunath. *GPS-MTM: Capturing Pattern of Normalcy in GPS-Trajectories with Self-Supervised Learning*. arXiv:2509.24031, 2025.
- [2] **First author.** Umang Garg, et al. *Symmetry-aware super-resolution of crystal orientation maps via invariant latent-space learning*. Manuscript under review at *Nature Machine Intelligence*.
- [3] Dr. Chen, Umang Garg, et al. *Automated Neural Network Quantification of Ventricular Volume Change on Routine CT After Shunting for Idiopathic Normal Pressure Hydrocephalus*. Manuscript under review at *Neurosurgery Journal*.

Published Papers

- [4] **First author.** Umang Garg, Kezhou Yang, and Abhronil Sengupta. *Emulation of Astrocyte Induced Neural Phase Synchrony in Spin-Orbit Torque Oscillator Neurons*. *Frontiers in Neuroscience*, 15:699632, 2021. doi: 10.3389/fnins.2021.699632.
- [5] **First author.** Umang Garg, Nithin Rahav J. K., and B. Sainath. *Spectrally Efficient Cooperative Visible Light Communication with Adaptive Power Sharing for a Generalized System*. In *2020 29th Wireless and Optical Communications Conference (WOCC)*, pp. 1–6, 2020. doi: 10.1109/WOCC48579.2020.9114946.
- [6] Nishant Gupta, Umang Garg, Sakshi Agarwal, and Mohit Vyas. *Onboard and Ground Station Telemetry Architecture Design for a LEO Nanosatellite*. In *2020 IEEE Aerospace Conference*, pp. 1–18, 2020. doi: 10.1109/AERO47225.2020.9172474.
- [7] **First author.** Umang Garg and Rutwik N. Jain. *Modelling and Simulation of Onboard Wire Antennas for a 3U CubeSat*. In *2019 Photonics & Electromagnetics Research Symposium – Fall (PIERS – Fall)*, 2019. doi: 10.1109/PIERS-Fall48861.2019.9021446.

Patents

- [8] *Method and System for Controlling a Qubit for Obtaining a Scalable Structure of a Hybrid Quantum-Classical Architecture*. US Patent Application Publication US20220398484A1, 2022.
- [9] *Method and System for Generating and Regulating Local Magnetic Field Variations for Spin Qubit Manipulation Using Micro-Structures in Integrated Circuits*. US Patent Application Publication US20220269971A1, 2022.

INDUSTRY EXPERIENCE

Mentium Technologies  | Machine-Learning Intern

Aug ‘23 – Sep ‘23 | Santa Barbara, CA

- Developing front-end framework for uploaded ML model evaluation on in-house ML accelerator.

QpiAI Technologies  | Design Engineer Intern

Dec ‘20 – Jul ‘21 | Bangalore, India

- Developed a scalable in-silico solution for magnetic-field control of qubits; lead inventor on **two U.S. patent applications**.
- Filed a **U.S. patent application** proposing an auxiliary pulse-control mechanism that improved in-silicon fidelity by $10\times$.

RELEVANT COURSEWORK

Digital Image Processing, Machine Intelligence, Machine Learning, Data Structures and Algorithms, Adversarial Machine Learning, Advanced Computer Vision, Artificial Intelligence, NP-hard Optimizations, Deep-learning, Software-HW Co-Design, Embedded Systems.

RESEARCH AWARDS AND LEADERSHIP

- [1] Graduate Research Accelerator Fellowship Recipient, 2025
- [2] ECE Department Fellowship Recipient, 2026
- [3] Led Team "Reflexn" at the Venture Fair to develop automated surgical planning for facial surgery. [Poster](#)
- [4] Served on the Graduate Student Association Board for San Clemente Villages, representing 800+ graduate students
- [5] Taught Digital Image Processing (ECE 178), Computer Vision (ECE 181), ECE 10B, and ECE 3 at the university